Computer Communications 154 (2020) 11

Contents lists available at ScienceDirect

Computer Communications

journal homepage: www.elsevier.com/locate/comcom

Editorial

IFIP Networking 2018 Special Issue

This Special Issue of Elsevier's Computer Communications collects the best papers of the 2018 edition of IFIP Networking, which was held in Zurich, Switzerland. After a thorough reviewing process that lead to the acceptance of 55 papers out of 225 submitted works, the 10 papers that received the highest scores by reviewers were invited to submit an extended version for a journal publication. This first selection used timeliness, contribution and quality of technical results presented, as the major criteria that warranted the invitation.

Beside taking into account all suggestions and recommendations given by reviewers, extended versions of the conference papers were expected to provide one or more of the following:

- Details on methods used in view of facilitating the reproducibility of results
- Information about any software used to obtain the results
- Data used for producing the results
- Results in the form of documented and machine-processable data sets

Revised, extended manuscripts were thus submitted for further review, involving both the same reviewers from the original conference submission as well as other highly qualified reviewers in order to guarantee the highest quality of the papers finally accepted for publication on Computer Communications.

The Guest Editors oversaw the whole process to guarantee that a rigorous, fair selection resulted in the highest possible quality of the present Special Issue.

As a result of the review process, 7 papers were eventually accepted for the Special Issue. They cover a wide variety of research areas and topics.

In "SRv6Pipes: enabling in-network bytestream functions", the authors explore the traffic engineering capabilities of IPv6 Segment Routing in a linux-based implementation.

"Prioritize Efficiently: Layer Selection in Network Coding" proposes a structured, layered network coding approach to minimize delay in sensor data exchange for smart-factory applications.

The authors of "Dynamic Load Balancing with Tokens" propose a token-based algorithm to agnostically balance the load on Data Center servers.

The evaluation of caching policies for various performance metrics in a hierarchical network of caches is the subject of "Performance Estimation and Evaluation Framework for Caching Policies in Hierarchical Caches".

In "Machine Learning for Measurement-based Bandwidth Estimation", packet dispersion is used to train a neural network and compare its effectiveness in estimating the bandwidth with other state-of-the-art techniques. The authors of "Reproducible Measurements of TCP BBR Congestion Control" present a framework to analyze different congestion control algorithms using network emulation, with a focus on Google's TPC BBR protocol.

Finally, Network Calculus is the tool used by "h-Mitigators: Improving Your Stochastic Network Calculus Output Bounds" to provide output delay bounds of cross-traffic flows.

We hope readers will find this selection of papers useful for their ongoing work as well as of inspiration to open new avenues of research. At the same time, we expect that Networking 2018 attendees will have the opportunity to gain additional insights on the best papers presented at the conference.

We wish to thank the Editor-in-Chief of Computer Communications, Carla Fabiana Chiasserini, and the Vice-Editor-in-Chief, Falko Dressler, for agreeing to this Special Issue. Likewise, we are also grateful to the Elsevier editorial team for speeding up the publication process.

To conclude this guest editorial, we wish to pay tribute to James Sterbenz, one of the Guest Editors, who sadly passed away a few months before the publication of the Special Issue, but who greatly contributed to the whole process from its inception.

This Special Issue is dedicated to his memory.

Claudio Casetti *
E-mail address: casetti@polito.it.

Jordi Domingo-Pascual

Fernando Kuipers

James Sterbenz

* Corresponding editor.

https://doi.org/10.1016/j.comcom.2020.02.022

Available online 7 February 2020
0140-3664/© 2020 Elsevier B.V. All rights reserved.